

**DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY
UTTAR PRADESH, LUCKNOW**



Evaluation Scheme & Syllabus

For

B.Tech. 3rd Year

COMPUTER SCIENCE AND ENGINEERING (INTERNET OF THINGS)

Based On

NEP2020

(Effective from the Session: 2024-25)

COMPUTER SCIENCE AND ENGINEERING (INTERNET OF THINGS) CURRICULUM STRUCTURE

**The Mini Project or internship (4 weeks) conducted during summer break after IV semester and will be assessed during V semester.

**It is desirable that the students should do their Summer Internship and Minor Project in their specialization area in line with the B.Tech program.

SEMESTER- VI													
Sl. No.	Subject Codes	Subject	Periods			Evaluation Scheme				End Semester		Total	Credit
			L	T	P	CT	TA	Total	PS	TE	PE		
1	BCS601	Software Engineering	3	1	0	20	10	30		70		100	4
2	BCIT601	Data Science in IOT	3	1	0	20	10	30		70		100	4
3	BCS603	Computer Networks	3	1	0	20	10	30		70		100	4
4	BCIT061/ BCDS062/ BCS063/ BCS064	Departmental Elective-III	3	0	0	20	10	30		70		100	3
5		Open Elective-I	3	0	0	20	10	30		70		100	3
6	BCS651	Software Engineering Lab	0	0	2				50		50	100	1
7	BCIT651	Internet of Things Lab	0	0	2				50		50	100	1
8	BCS653	Computer Networks Lab	0	0	2				50		50	100	1
9	BNC601/ BNC602	Constitution of India/ Essence of Indian Traditional Knowledge	2	0	0	20	10	30		70			
		Total	17	3	6							800	21
Minor Degree/Honors Degree MT-1/HT-1													
*The Mini Project or internship (4 weeks) will be done during summer break after 6th Semester and will be assessed during 7th semester.													
*It is desirable that the students should do their Summer Internship and Minor Project in their specialization area in line with the B.Tech. program.													

Departmental Elective-I

1. BCIT051 Mobile App Development for IoT
2. BCS052 Data Analytics
3. BCS053 Computer Graphics
4. BCIT052 IoT Architecture and Protocols

Departmental Elective-II

1. BCIT053 Computer Vision
2. BCIT054 Artificial Intelligence
3. BCIT055 Programming and Interfacing with Microcontrollers
4. BCIT056 Privacy and Security in IoT

Departmental Elective-III

1. BCIT061 Energy Harvesting Technologies and Power Management for IoT Devices
2. BCDS062 Machine Learning Techniques
3. BCS063 Blockchain Architecture Design
4. BCS064 Data Compression

B.TECH.
COMPUTER SCIENCE AND ENGINEERING (INTERNET OF THINGS)
FIFTH SEMESTER (DETAILED SYLLABUS)

DATABASE MANAGEMENT SYSTEM (BCS501)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to understand		
CO 1	Apply knowledge of database for real life applications.	K ₃
CO 2	Apply query-processing techniques to automate the real time problems of databases.	K ₃ , K ₄
CO 3	Identify and solve the redundancy problem in database tables using normalization.	K ₂ , K ₃
CO 4	Understand the concepts of transactions, their processing so they will familiar with broad range of database management issues including data integrity, security and recovery.	K ₂ , K ₄
CO 5	Design, develop and implement a small database project using database tools.	K ₃ , K ₆
DETAILED SYLLABUS		3-1-0
Unit	Topic	Proposed Lecture
I	Introduction: Overview, Database System vs File System, Database System Concept and Architecture, Data Model Schema and Instances, Data Independence and Database Language and Interfaces, Data Definitions Language, DML, Overall Database Structure. Data Modeling Using the Entity Relationship Model: ER Model Concepts, Notation for ER Diagram, Mapping Constraints, Keys, Concepts of Super Key, Candidate Key, Primary Key, Generalization, Aggregation, Reduction of an ER Diagrams to Tables, Extended ER Model, Relationship of Higher Degree.	08
II	Relational data Model and Language: Relational Data Model Concepts, Integrity Constraints, Entity Integrity, Referential Integrity, Keys Constraints, Domain Constraints, Relational Algebra, Relational Calculus, Tuple and Domain Calculus. Introduction on SQL: Characteristics of SQL, Advantage of SQL. SQL Data Type and Literals. Types of SQL Commands. SQL Operators and Their Procedure. Tables, Views and Indexes. Queries and Sub Queries. Aggregate Functions. Insert, Update and Delete Operations, Joins, Unions, Intersection, Minus, Cursors, Triggers, Procedures in SQL/PL SQL	08
III	Data Base Design & Normalization: Functional dependencies, normal forms, first, second, 8 third normal forms, BCNF, inclusion dependence, loss less join decompositions, normalization using FD, MVD, and JDs, alternative approaches to database design	08
IV	Transaction Processing Concept: Transaction System, Testing of Serializability, Serializability of Schedules, Conflict & View Serializable Schedule, Recoverability, Recovery from Transaction Failures, Log Based Recovery, Checkpoints, Deadlock Handling. Distributed Database: Distributed Data Storage, Concurrency Control, Directory System.	08
V	Concurrency Control Techniques: Concurrency Control, Locking Techniques for Concurrency Control, Time Stamping Protocols for Concurrency Control, Validation Based Protocol, Multiple Granularity, Multi Version Schemes, Recovery with Concurrent Transaction, Case Study of Oracle.	08
Text books: 1. Korth, Silbertz, Sudarshan," Database Concepts", McGraw Hill 2. Date C J, "An Introduction to Database Systems", Addison Wesley 3. Elmasri, Navathe, " Fundamentals of Database Systems", Addison Wesley 4. O'Neil, Databases, Elsevier Pub. 5. RAMAKRISHNAN"Database Management Systems", McGraw Hill 6. Leon & Leon,"Database Management Systems", Vikas Publishing House 7. Bipin C. Desai, " An Introduction to Database Systems", Gagotia Publications 8. Majumdar & Bhattacharya, "Database Management System", TMH		

WEB TECHNOLOGY (BCS 502)

Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to		
CO 1	Understand the fundamental concepts of web development, including the history, protocols, and tools. Apply HTML and XML in the development of web projects.	K ₃ , K ₆
CO 2	Apply CSS for designing and styling web pages, including the use of CSS properties, styling elements, and advanced techniques for creating responsive web sites.	K ₂ , K ₃
CO 3	Develop interactive web applications using JavaScript and AJAX, with a focus on scripting documents, forms, and networking concepts such as internet addressing and TCP/IP sockets.	K ₃ , K ₆
CO 4	Design and implement server-side applications using Enterprise Java Beans (EJB) and Node.js, including the creation of JavaBeans, RESTful APIs, and database operations with MongoDB	K ₂ , K ₄ , K ₆
CO 5	Implement web server functionality using Servlets and Java Server Pages (JSP), focusing on handling HTTP requests, session tracking, and utilizing custom tag libraries for dynamic web content.	K ₂ , K ₃ , K ₄
CO 1	Understand the fundamental concepts of web development, including the history, protocols, and tools. Apply HTML and XML in the development of web projects.	K ₃ , K ₆
DETAILED SYLLABUS		3-1-0
Unit	Topic	Proposed Lecture
I	Introduction: Introduction and Web Development Strategies, History of Web and Internet, Protocols Governing Web, Writing Web Projects, Connecting to Internet, Introduction to Internet services and tools, Introduction to client-server computing. Web Page Designing: HTML: List, Table, Images, Frames, forms, XML: Document type definition (DTD), XML schemes, Object Models, presenting and using XML, Using XML Processors: DOM and SAX.	08
II	CSS: Creating Style Sheet, CSS Properties, CSS Styling (Background, Text Format, Controlling Fonts), Working with block elements and objects, Working with Lists and Tables, CSS Id and Class, Box Model (Introduction, Border properties, Padding Properties, Margin properties) CSS Advanced (Grouping, Dimension, Display, Positioning, Floating, Align, Pseudo class, Navigation Bar, Image Sprites, Attribute selector), CSS Color, Creating page Layout and Site Designs.	08
III	Scripting: Java script: Introduction, documents, forms, statements, functions, objects, introduction to AJAX. Networking: Internet Addressing, InetAddress, Factory Methods, Instance Methods, TCP/IP Client Sockets, URL, URL Connection, TCP/IP Server Sockets, Datagram.	08
IV	Enterprise Java Bean: Creating a JavaBeans, JavaBeans Properties, Types of beans, Stateful Session bean, Stateless Session bean, Entity bean. Node.js: Introduction, Environment Setup, REPL Terminal, NPM (Node Package Manager) Callbacks Concept, Events, Packaging, Express Framework, Restful API. Node.js with MongoDB: MongoDB Create Database, Create Collection, Insert, delete, update, join, sort, query.	08
V	Servlets: Servlet Overview and Architecture, Interface Servlet and the Servlet Life Cycle, Handling HTTP get Requests, Handling HTTP post Requests, Redirecting Requests to Other Resources, Session Tracking, Cookies, Session Tracking with Http Session Java Server Pages (JSP): Introduction, Java Server Pages Overview, A First Java Server Page Example, Implicit Objects, Scripting, Standard Actions, Directives, Custom Tag Libraries	08

Text books:

1. Burdman, Jessica, "Collaborative Web Development" Addison Wesley
2. Xavier, C, "Web Technology and Design", New Age International
3. Ivan Bayross, "HTML, DHTML, Java Script, Perl & CGI", BPB Publication
4. Hans Bergsten, "Java Server Pages", SPD O'Reilly
5. Margaret Levine Young, "The Complete Reference Internet", McGraw Hill.
6. Greg Lim, "Beginning Node.js, Express & MongoDB Development", 1 September 2020, Greg Lim
7. Shannon Bradshaw, Eoin Brazil, Kristina Chodorow, "MongoDB: The Definitive Guide, 3rd Edition", December 2019, O'Reilly Media, Inc.

DESIGN AND ANALYSIS OF ALGORITHM (BCS 503)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to understand		
CO 1	Design new algorithms, prove them correct, and analyze their asymptotic and absolute runtime and memory demands.	K ₄ , K ₆
CO 2	Find an algorithm to solve the problem (create) and prove that the algorithm solves the problem correctly (validate).	K ₅ , K ₆
CO 3	Understand the mathematical criterion for deciding whether an algorithm is efficient, and know many practically important problems that do not admit any efficient algorithms.	K ₂ , K ₅
CO 4	Apply classical sorting, searching, optimization and graph algorithms.	K ₂ , K ₄
CO 5	Understand basic techniques for designing algorithms, including the techniques of recursion, divide-and-conquer, and greedy.	K ₂ , K ₃
DETAILED SYLLABUS		3-1-0
Unit	Topic	Proposed Lecture
I	Introduction: Algorithms, Analyzing Algorithms, Complexity of Algorithms, Growth of Functions, Performance Measurements, Sorting and Order Statistics - Shell Sort, Quick Sort, Merge Sort, Heap Sort, Comparison of Sorting Algorithms, Sorting in Linear Time.	08
II	Advanced Data Structures: Red-Black Trees, B – Trees, Binomial Heaps, Fibonacci Heaps, Tries, Skip List	08
III	Divide and Conquer with Examples Such as Sorting, Matrix Multiplication, Convex Hull and Searching. Greedy Methods with Examples Such as Optimal Reliability Allocation, Knapsack, Minimum Spanning Trees – Prim's and Kruskal's Algorithms, Single Source Shortest Paths - Dijkstra's and Bellman Ford Algorithms.	08
IV	Dynamic Programming with Examples Such as Knapsack. All Pair Shortest Paths – Warshal's and Floyd's Algorithms, Resource Allocation Problem. Backtracking, Branch and Bound with Examples Such as Travelling Salesman Problem, Graph Coloring, n-Queen Problem, Hamiltonian Cycles and Sum of Subsets.	08
V	Selected Topics: Algebraic Computation, Fast Fourier Transform, String Matching, Theory of NP-Completeness, Approximation Algorithms and Randomized Algorithms	08
Text books: 1. Thomas H. Coreman, Charles E. Leiserson and Ronald L. Rivest, “Introduction to Algorithms”, Printice Hall of India. 2. E. Horowitz & S Sahni, "Fundamentals of Computer Algorithms", 3. Aho, Hopcraft, Ullman, “The Design and Analysis of Computer Algorithms” Pearson Education, 2008. 4. LEE "Design & Analysis of Algorithms (POD)", McGraw Hill 5. Richard E. Neapolitan "Foundations of Algorithms" Jones & Bartlett Learning 6. Jon Kleinberg and Éva Tardos, Algorithm Design, Pearson, 2005. 7. Michael T Goodrich and Roberto Tamassia, Algorithm Design: Foundations, Analysis, and Internet Examples, Second Edition, Wiley, 2006. 8. Harry R. Lewis and Larry Denenberg, Data Structures and Their Algorithms, Harper Collins, 1997 9. Robert Sedgewick and Kevin Wayne, Algorithms, fourth edition, Addison Wesley, 2011. 10. Harsh Bhasin, "Algorithm Design and Analysis", First Edition, Oxford University Press. 11. Gilles Brassard and Paul Bratley, Algorithmics: Theory and Practice, Prentice Hall, 1995.		

Mobile Application Development for IoT (BCIT051)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able:		
CO 1	Able to understand the application areas of IOT	K ₁ , K ₂
CO 2	Able to realize the revolution of Internet in Mobile Devices, Cloud & Sensor Networks	K ₁ , K ₃
CO 3	Able to understand building blocks of Internet of Things and characteristics	K ₂ , K ₃
CO 4	Understand the concepts of mobile apps used in internet of things	K ₄ , K ₆
CO 5	Learn about android application development	K ₄ , K ₅
CO6	Implement mobile application using Android things	
DETAILED SYLLABUS		3-0-0
Unit	Topic	Proposed Lecture
I	UNIT I OVERVIEW IoT -An Architectural Overview Building an architecture, Main design principles and needed capabilities, An IoT architecture outline, standards considerations. M2M and IoT Technology Fundamentals- Devices and gateways, Local and wide area networking, Data management, Business processes in IoT, Everything as a Service(XaaS), M2M and IoT Analytics, Knowledge Management;	08
II	UNIT II BASIC DESIGN Introduction Basics of embedded systems design Embedded OS - Design constraints for mobile applications, both hardware and software related Architecting mobile applications user interfaces for mobile applications touch events and gestures Achieving quality constraints performance, usability, security, availability and modifiability.	08
III	Unit III IoT MOBILE APPS IoT Mobile App Development Trends In 2020 - Role of Mobile Apps in revolutionizing the world of IoT - UX / UI design for IoT Mobile apps - challenges of UX/UI design for IoT applications - practice tips on design for IoT mobile apps IoT App Design Solutions	08
IV	UNIT IV TECHNOLOGY I- ANDROID Introduction Establishing the development environment Android architecture Activities and views Interacting with UI Persisting data using SQLite Packaging and deployment Interaction with server side applications Using Google Maps, GPS and Wifi Integration with social media applications.	08
V	UNIT V TECHNOLOGY II ANDROID THINGS Android things Installation of Android things Verification of Android things installation API in Android IoT Setting up the Android things - Building your first IOT-Android application with Android Things	08
Text books: 1. Bernd Scholz- -3-642-19156-5 e-ISBN 978-3- 642-19157-2, Springer 2. Jeff McWherter and Scott Gowell, "Professional Mobile Application Development", Wrox, 2012		

Data Analytics (BCS052)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to :		
CO 1	Describe the life cycle phases of Data Analytics through discovery, planning and building.	K1,K2
CO 2	Understand and apply Data Analysis Techniques.	K2, K3
CO 3	Implement various Data streams.	K3
CO 4	Understand item sets, Clustering, frame works & Visualizations.	K2
CO 5	Apply R tool for developing and evaluating real time applications.	K3,K5,K6
DETAILED SYLLABUS		3-0-0
Unit	Topic	Proposed Lecture
I	Introduction to Data Analytics: Sources and nature of data, classification of data (structured, semi-structured, unstructured), characteristics of data, introduction to Big Data platform, need of data analytics, evolution of analytic scalability, analytic process and tools, analysis vs reporting, modern data analytic tools, applications of data analytics. Data Analytics Lifecycle: Need, key roles for successful analytic projects, various phases of data analytics lifecycle – discovery, data preparation, model planning, model building, communicating results, operationalization.	08
II	Data Analysis: Regression modeling, multivariate analysis, Bayesian modeling, inference and Bayesian networks, support vector and kernel methods, analysis of time series: linear systems analysis & nonlinear dynamics, rule induction, neural networks: learning and generalisation, competitive learning, principal component analysis and neural networks, fuzzy logic: extracting fuzzy models from data, fuzzy decision trees, stochastic search methods.	08
III	Mining Data Streams: Introduction to streams concepts, stream data model and architecture, stream computing, sampling data in a stream, filtering streams, counting distinct elements in a stream, estimating moments, counting oneness in a window, decaying window, Real-time Analytics Platform (RTAP) applications, Case studies – real time sentiment analysis, stock market predictions.	08
IV	Frequent Itemsets and Clustering: Mining frequent itemsets, market based modelling, Apriori algorithm, handling large data sets in main memory, limited pass algorithm, counting frequent itemsets in a stream, clustering techniques: hierarchical, K-means, clustering high dimensional data, CLIQUE and ProCLUS, frequent pattern based clustering methods, clustering in non-euclidean space, clustering for streams and parallelism.	08
V	Frame Works and Visualization: MapReduce, Hadoop, Pig, Hive, HBase, MapR, Sharding, NoSQL Databases, S3, Hadoop Distributed File Systems, Visualization: visual data analysis techniques, interaction techniques, systems and applications. Introduction to R - R graphical user interfaces, data import and export, attribute and data types, descriptive statistics, exploratory data analysis, visualization before analysis, analytics for unstructured data.	08
Text books and References:		
1. Michael Berthold, David J. Hand, Intelligent Data Analysis, Springer 2. Anand Rajaraman and Jeffrey David Ullman, Mining of Massive Datasets, Cambridge University Press. 3. Bill Franks, Taming the Big Data Tidal wave: Finding Opportunities in Huge Data Streams with Advanced Analytics, John Wiley & Sons. 4. John Garrett, Data Analytics for IT Networks : Developing Innovative Use Cases, Pearson Education		

Computer Graphics (BCS053)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to:		
CO 1	Understand the graphics hardware used in field of computer graphics.	K ₂
CO 2	Understand the concept of graphics primitives such as lines and circle based on different algorithms.	K ₂ , K ₄
CO 3	Apply the 2D graphics transformations, composite transformation and Clipping concepts.	K ₄
CO 4	Apply the concepts of and techniques used in 3D computer graphics, including viewing transformations.	K ₂ , K ₃
CO 5	Perform the concept of projections, curve and hidden surfaces in real life.	K ₂ , K ₃
DETAILED SYLLABUS		3-0-0
Unit	Topic	Proposed Lecture
I	Introduction and Line Generation: Types of computer graphics, Graphic Displays- Random scan displays, Raster scan displays, Frame buffer and video controller, Points and lines, Line drawing algorithms, Circle generating algorithms, Mid-point circle generating algorithm, and parallel version of these algorithms.	08
II	Transformations: Basic transformation, Matrix representations and homogenous coordinates, Composite transformations, Reflections and shearing. Windowing and Clipping: Viewing pipeline, Viewing transformations, 2-D Clipping algorithms- Line clipping algorithms such as Cohen Sutherland line clipping algorithm, Liang Barsky algorithm, Line clipping against non rectangular clip windows; Polygon clipping – Sutherland Hodgeman polygon clipping, Weiler and Atherton polygon clipping, Curve clipping, Text clipping	08
III	Three Dimensional: 3-D Geometric Primitives, 3-D Object representation, 3-D Transformation, 3-D viewing, projections, 3-D Clipping.	08
IV	Curves and Surfaces: Quadric surfaces, Spheres, Ellipsoid, Blobby objects, Introductory concepts of Spline, Bspline and Bezier curves and surfaces.	08
V	Hidden Lines and Surfaces: Back Face Detection algorithm, Depth buffer method, A- buffer method, Scan line method, basic illumination models– Ambient light, Diffuse reflection, Specular reflection and Phong model, Combined approach, Warn model, Intensity Attenuation, Color consideration, Transparency and Shadows.	08
Text books: 1. Donald Hearn and M Pauline Baker, “Computer Graphics C Version”, Pearson Education 2. Foley, Vandom, Feiner, Hughes – “Computer Graphics principle”, Pearson Education. 3. Rogers, “ Procedural Elements of Computer Graphics”, McGraw Hill 4. W. M. Newman, R. F. Sproull – “Principles of Interactive computer Graphics” – McGraw Hill. 5. Amrendra N Sinha and Arun D Udai,” Computer Graphics”, McGraw Hill. 6. R.K. Maurya, “Computer Graphics ” Wiley Dreamtech Publication. 7. Mukherjee, Fundamentals of Computer graphics & Multimedia, PHI Learning Private Limited. 8. Donald Hearn and M Pauline Baker, “Computer Graphics with Open GL”, Pearson education		

IOT ARCHITECTURE AND PROTOCOLS(BCIT052)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to		
CO 1	To Understand the Architectural Overview of IoT	K ₁ ,K ₂
CO 2	To Understand the IoT Reference Architecture and RealWorldDesign Constraints	K ₁ ,K ₂
CO 3	To Understand the various IoT Protocols (Datalink, Network, Transport, Session, Service)	K ₃
CO 4	Application of various IoT Protocols (Datalink, Network, Transport, Session, Service)	K ₄ ,K ₅
CO 5	To Understand the various Real-World Design Constraints.	K ₄ ,K ₅
DETAILED SYLLABUS		3-0-0
Unit	Topic	Proposed Lectures
I	IoT-An Architectural Overview: Building an architecture, Main design principles and needed capabilities, An IoT architecture outline, standards considerations. M2M and IoT Technology Fundamentals- Devices and gateways, Local and wide area networking, Data management, Business processes in IoT, Everything as a Service(XaaS), M2M and IoT Analytics, Knowledge Management	06
II	Reference Architecture: IoT Architecture-State of the Art – Introduction, State of the art, Reference Model and architecture, IoT reference Model - IoT Reference Architecture- Introduction,Functional View, Information View, Deployment and Operational View, Other Relevant architectural views. Real-World Design Constraints- Introduction, Technical Design constraints- hardware is popular again, Data representation and visualization, Interaction and remote control.	08
III	IOT Data Link Layer & Network Layer Protocols : PHY/MAC Layer(3GPP MTC, IEEE 802.11, IEEE 802.15), WirelessHART, ZWave, Bluetooth Low Energy, Zigbee Smart Energy,DASH7 - Network Layer-IPv4, IPv6, 6LoWPAN, 6TiSCH,ND, DHCP, ICMP, RPL, CORPL, CARP	08
IV	Transport & Session Layer Protocols: Transport Layer (TCP, MPTCP, UDP, DCCP, SCTP)-(TLS, DTLS) – Session Layer-HTTP, CoAP, XMPP, AMQP, MQTT	08
V	Service Layer Protocols & Security : Service Layer -oneM2M, ETSI M2M, OMA, BBF – Security in IoT Protocols – MAC 802.15.4 , 6LoWPAN, RPL, Application Layer	08
Text books 1. Jan Holler, VlasiosTsiatsis, Catherine Mulligan, Stefan Avesand,StamatisKarnouskos, David Boyle, “From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence”, 1st Edition, Academic Press,2014. 2. Peter Waher, “Learning Internet of Things”, PACKT publishing, BIRMINGHAM –MUMBAI 3. Bernd Scholz-Reiter, Florian Michahelles, “Architecting the Internet of Things”,ISBN 978-3-642-19156-5 e-ISBN 978-3-642-19157-2, Springer 4. Daniel Minoli, “Building the Internet of Things with IPv6 and MIPv6: The Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications 5. Vijay Madisetti and ArshdeepBahga, “Internet of Things (A Hands-on-Approach)”, 1st Edition, VPT, 2014.		

COMPUTER VISION(BCIT053)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to:		
CO 1	Implement fundamental image processing techniques required for computer vision.	K ₁ , K ₂
CO 2	Perform shape analysis and implement boundary tracking techniques.	K ₁ , K ₂
CO 3	Apply Hough Transformation for line, circle, ellipse detections.	K ₃ , K ₄
CO 4	Apply 3D vision and motion related techniques.	K ₃ , K ₄
CO 5	Develop applications using Computer vision techniques.	K ₁ , K ₂
DETAILED SYLLABUS		3-0-0
Unit	Topic	Proposed Lecture
I	1. Image Processing Fundamentals: Review of image processing techniques, Classical filtering operations, Thresholding Techniques, Edge detection techniques, Corner and interest point detection, Mathematical Morphology, Texture.	08
II	1. Shapes and Regions: Binary shape analysis, connectedness, object labeling and counting, size filtering, distance functions, Skeletons and thinning, deformable shape analysis, Boundary tracking procedure, active contours, Shape model and shape recognition, centroidal profiles, handling occlusion, boundary length measure, boundary descriptors, chain codes, Fourier descriptors, region descriptors.	08
III	1. Hough Transform: Line detection, Hough Transform (HT) for line detection, foot of normal method, line localization, line fitting, RANSAC for straight line detection, HT based circular object detection, accurate center location, speed problem, ellipse detection, Case Studies: Human Iris location, hole detection, generalized Hough Transformation (GHT), Spatial matched filtering, GHT for ellipse detection, object location, GHT for feature collation.	08
IV	Vision and Motion: Methods for 3D vision, projection schemes, shape from shading, Photometric stereo, shape from texture, shape from focus, active range finding, surface representation, point-based representation, Volumetric representation, 3D object recognition, 3D reconstruction. Introduction to motion, triangulation, bundle adjustment, translational alignment, parametric motion, Spline-based motion, optional flow, layered motion.	08
V	Applications: Photo Album: Face detection, face recognition, Eigen faces, Active appearance and 3D shape models for face applications. Surveillance: Foreground and background separation, Particle filters, Chamfer matching, tacking and occlusion, combining view from multiple cameras, Human gait analysis. In vehicle vision system: locating roadways, road marking, identifying road signs, locating pedestrians.	08
TEXT BOOKS		
1. E. R. Davies, Computer and Machine Vision, Forth Edition, Academic Press, 2011 2. R. Szeliski, Computer Vision: Algorithms and Applications, Springer, 2011 3. Simon D. J. Prince, Computer Vision: Models, Learning, and Inference, Cambridge University Press, 2012.		
REFERENCE BOOKS		
1. D. L. Baggio et.al., Mastering OpenCV with Practical computer vision projects, Packt publishing, 2012. 2. Jan Erik Solem, Programming Computer Vision with Python: Tools and Algorithms for analysing images, O'Reilly Media, 2012 3. Mark Nixon and Alberto S Aquado, Feature extraction and image processing for computer vision, Third edition, Academic Press, 2012		

ARTIFICIAL INTELLIGENCE (BCIT054)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to understand		
CO 1	Understand the basics of the theory and practice of Artificial Intelligence as a discipline and about intelligent agents.	K ₂
CO 2	Understand search techniques and gaming theory.	K ₂ , K ₃
CO 3	The student will learn to apply knowledge representation techniques and problem solving strategies to common AI applications.	K ₃ , K ₄
CO 4	Student should be aware of techniques used for classification and clustering.	K ₂ , K ₃
CO 5	Student should aware of basics of pattern recognition and steps required for it.	K ₂ , K ₄
DETAILED SYLLABUS		3-0-0
Unit	Topic	Proposed Lecture
I	INTRODUCTION : Introduction–Definition – Future of Artificial Intelligence – Characteristics of Intelligent Agents– Typical Intelligent Agents – Problem Solving Approach to Typical AI problems.	08
II	PROBLEM SOLVING METHODS Problem solving Methods – Search Strategies- Uninformed – Informed – Heuristics – Local Search Algorithms and Optimization Problems – Searching with Partial Observations – Constraint Satisfaction Problems – Constraint Propagation – Backtracking Search – Game Playing – Optimal Decisions in Games – Alpha – Beta Pruning – Stochastic Games	08
III	KNOWLEDGE REPRESENTATION First Order Predicate Logic – Prolog Programming – Unification – Forward Chaining-Backward Chaining – Resolution – Knowledge Representation – Ontological Engineering-Categories and Objects – Events – Mental Events and Mental Objects – Reasoning Systems for Categories – Reasoning with Default Information	08
IV	SOFTWARE AGENTS Architecture for Intelligent Agents – Agent communication – Negotiation and Bargaining – Argumentation among Agents – Trust and Reputation in Multi-agent systems.	08
V	APPLICATIONS AI applications – Language Models – Information Retrieval- Information Extraction – Natural Language Processing – Machine Translation – Speech Recognition – Robot – Hardware – Perception – Planning – Moving	08
Text books: 1. S. Russell and P. Norvig, “Artificial Intelligence: A Modern Approach”, Prentice Hall, Third Edition, 2009. 2. I. Bratko, —Prolog: Programming for Artificial Intelligencel, Fourth edition, Addison-Wesley Educational Publishers Inc., 2011. 3. M. Tim Jones, —Artificial Intelligence: A Systems Approach(Computer Science)l, Jones and Bartlett Publishers, Inc.; First Edition, 2008 4. Nils J. Nilsson, —The Quest for Artificial Intelligencel, Cambridge University Press, 2009. 5. William F. Clocksin and Christopher S. Mellish,l Programming in Prolog: Using the ISO Standardl, Fifth Edition, Springer, 2003. 6. Gerhard Weiss, —Multi Agent Systemsll, Second Edition, MIT Press, 2013. 7. David L. Poole and Alan K. Mackworth, —Artificial Intelligence: Foundations of Computational Agentsl, Cambridge University Press, 2010.		

PROGRAMMING AND INTERFACING WITH MICROCONTROLLERS (BCIT055)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to:		
CO 1	Understand advanced and emerging networking technologies	K ₁ , K ₂
CO 2	Obtain skills to do advanced networking research and programming	K ₃ , K ₄
CO 3	Learn how to use software programs to perform varying and complex networking tasks	K ₁ , K ₂
CO 4	Understand Advanced I/O Interfacing	K ₂ , K ₃
CO 5	Expand upon the knowledge learned and apply it to solve real world problems	K ₅ , K ₆
DETAILED SYLLABUS		3-0-0
Unit	Topic	Proposed Lecture
I	INTRODUCTION Introduction – History – Creative Coding Platforms – Open Source Platforms – PIC – Arduino, Sketch, Raspberry Pi, Iterative coding methodology – Python Programming – Mobile phones and similar devices – Arm Devices – Getting used to Arduino – Sensor Characterization: Safety, Basic Electronics (circuit theory, measurements, parts identification) Sensors and Software: Understanding Processing Code Structure, variables and flow control, Interfacing to the Real World	08
II	SOFTWARE FRAMEWORKS Software: openFrameworks as our IDE (C/C++) – “Arduino” Language (C/C++) – Hardware: Desktop / Laptop / Raspberry Pi – How to approach a programming problem? Representing “reality” with computers. Digital vs. Analog circuits, audio, communication, etc. Analog to Digital Conversion (ADC) – Digital to Analog Conversion (DAC) – Microcontrollers	08
III	HARDWARE COMMUNICATION Communication – Serial & Parallel – Hardware to Hardware Communication – I2C/IIC (Inter-Integrated Circuit) – SPI (Serial Peripheral Interface) – Serial UART Communication – Introduction to the command line – git/github. Introduction to Programming: A comparative studio between Arduino + open Frameworks – Arduino compatible Microcontrollers Sensors and Actuators	08
IV	ADVANCED I/O INTERFACING Advanced I/O – open Frameworks: Live Network feeds (push and pull) – Data persistence (saving data and preferences) – Database interface (MySQL, SQLite, XML, PHP/Web) – Arduino: Wired/Wireless Networking	08
V	IoT, FUTURE AND PERSPECTIVES Talking to the cloud: Baby steps to Internet of Things, TCP/IP and UDP – Building peer to peer communication system using Bluetooth & WiFi– Experiments	08
Text Books:		
1. Programming Interactivity, Second Edition By Josha Noble, 2012 2. Programming the Raspberry Pi: Getting Started with Python 2E, 2016		

PRIVACY AND SECURITY IN IOT(BCIT056)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to:		
CO 1	Ability to understand the Security requirements in IoT.	K ₁ , K ₂
CO 2	Understand the cryptographic fundamentals for IoT	K ₁ , K ₂
CO 3	Ability to understand the authentication credentials and access Control	K ₁ , K ₂
CO 4	Understand the various types Trust models and Cloud Security.	K ₁ , K ₂
DETAILED SYLLABUS		3-0-0
Unit	Topic	Proposed Lecture
I	INTRODUCTION: SECURING THE INTERNET OF THINGS: Security Requirements in IoT Architecture - Security in Enabling Technologies -Security Concerns in IoT Applications. Security Architecture in the Internet of Things -Security Requirements in IoT - Insufficient Authentication/Authorization – Insecure Access Control - Threats to Access Control, Privacy, and Availability - Attacks Specific to IoT. Vulnerabilities – Secrecy and Secret-Key Capacity - Authentication/Authorization for Smart Devices - Transport Encryption – Attack & Fault trees	08
II	CRYPTOGRAPHIC FUNDAMENTALS FOR IOT: Cryptographic primitives and its role in IoT – Encryption and Decryption – Hashes –Digital Signatures – Random number generation – Cipher suites – key management fundamentals – cryptographic controls built into IoT messaging and communication protocols – IoT Node Authentication	08
III	IDENTITY & ACCESS MANAGEMENT SOLUTIONS FOR IOT: Identity lifecycle – authentication credentials – IoT IAM infrastructure – Authorization with Publish / Subscribe schemes – access control	08
IV	PRIVACY PRESERVATION AND TRUST MODELS FOR IOT: Concerns in data dissemination – Lightweight and robust schemes for Privacy protection – Trust and Trust models for IoT – self-organizing Things - Preventing unauthorized access	08
V	CLOUD SECURITY FOR IOT: Cloud services and IoT – offerings related to IoT from cloud service providers – Cloud IoT security controls – An enterprise IoT cloud security architecture – New directions in cloud enabled IoT computing	08
Text Books: 1. Practical Internet of Things Security (Kindle Edition) by Brian Russell, Drew Van Duren 2. Securing the Internet of Things Elsevier 3. Security and Privacy in Internet of Things (IoTs): Models, Algorithms, and Implementations		

DATABASE MANAGEMENT SYSTEMS LAB (BCS551)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to:		
CO 1	Understand and apply oracle 11 g products for creating tables, views, indexes, sequences and other database objects.	K ₂ , K ₄
CO 2	Design and implement a database schema for company data base, banking data base, library information system, payroll processing system, student information system.	K ₃ , K ₅ , K ₆
CO 3	Write and execute simple and complex queries using DDL, DML, DCL and TCL	K ₄ , K ₅
CO 4	Write and execute PL/SQL blocks, procedure functions, packages and triggers, cursors.	K ₄ , K ₅
CO 5	Enforce entity integrity, referential integrity, key constraints, and domain constraints on database.	K ₃ , K ₄
DETAILED SYLLABUS		
1. Installing oracle/ MYSQL 2. Creating Entity-Relationship Diagram using case tools. 3. Writing SQL statements Using ORACLE /MYSQL: a) Writing basic SQL SELECT statements. b) Restricting and sorting data. c) Displaying data from multiple tables. d) Aggregating data using group function. e) Manipulating data. e) Creating and managing tables. 4. Normalization 5. Creating cursor 6. Creating procedure and functions 7. Creating packages and triggers 8. Design and implementation of payroll processing system 9. Design and implementation of Library Information System 10. Design and implementation of Student Information System 11. Automatic Backup of Files and Recovery of Files 12. Mini project (Design & Development of Data and Application) for following : a) Inventory Control System. b) Material Requirement Processing. c) Hospital Management System. d) Railway Reservation System. e) Personal Information System. f) Web Based User Identification System. g) Timetable Management System. h) Hotel Management System		
Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner It is also suggested that open source tools should be preferred to conduct the lab (MySQL , SQL server , Oracle ,MongoDB ,Cubrid ,MariaDBetc)		

Database Management Systems Lab BCS 551: Mapping with Virtual Lab

Name of the Lab	Name of the Experiment
Database Management Lab (BCS-551)	Data Definition Language(DDL) Statements: (Create table, Alter table, Drop table)
	Data Manipulation Language(DML) Statements
	Data Query Language(DQL) Statements: (Select statement with operations like Where clause, Order by, Logical operators, Scalar functions and Aggregate functions)
	Transaction Control Language(TCL) statements: (Commit(make changes permanent), Rollback (undo)
	Describe statement: To view the structure of the table created

Web Technology Lab (BCS552)

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to:

CO 1	Understanding fundamentals of website development and apply HTL and XML languages for development of websites	K ₂ , K ₄
CO 2	Applying CSS in designing and development of responsive website for compatibility of various devices.	K ₂ , K ₃ , K ₅
CO 3	Understand, analyze and design the role of JavaScript for dynamic web pages.	K ₂ , K ₄ , K ₅
CO 4	Design and deploy different components using Java Bean, Node.js and database tables using MongoDB and produce various results based on given query.	K ₄ , K ₅
CO 5	Design and deploy server-side java application called Servlet & JSP tools to catch form data sent from client, process it and store it on database.	K ₃ , K ₄

DETAILED SYLLABUS

This lab is based on the Web Technologies. Some examples are as follows:

1. Write HTML program for designing your institute website. Display departmental information of your institute on the website.
2. Write HTML program to design an entry form for student details/employee information/faculty details.
3. Develop a responsive website using CSS and HTML. Website may be for tutorial/blogs/commercial website.
4. Write programs using HTML and Java Script for validation of input data.
5. Write a program in XML for creation of DTD, which specifies set of rules. Create a style sheet in CSS/ XSL & display the document in internet explorer.
6. Create a Java Bean for Employee information (EmpID, Name, Salary, Designation and Department).
7. Build a command-line utility using Node.js that performs a specific task, such as converting text to uppercase, calculating the factorial of a number, or generating random passwords.
8. Develop a script that uses MongoDB's aggregation framework to perform operations like grouping, filtering, and sorting. For instance, aggregate user data to find the average age of users in different cities.
9. Assume four users user1, user2, user3 and user4 having the passwords pwd1, pwd2, pwd3 and pwd4 respectively. Write a servlet for doing the following: 1. Create a Cookie and add these four user id's and passwords to this Cookie. 2. Read the user id and passwords entered in the Login form and authenticate with the values available in the cookies.
10. Create a table which should contain at least the following fields: name, password, email-id, phone number. Write Servlet/JSP to connect to that database and extract data from the tables and display them. Insert the details of the users who register with the web site, whenever a new user clicks the submit button in the registration page.
11. Write a JSP which insert the details of the 3 or 4 users who register with the web site by using registration form. Authenticate the user when he submits the login form using the user name and password from the database.
12. Design and implement a simple shopping cart example with session tracking API.

**Note: The instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner
It is also suggested that open source tools should be preferred to conduct the lab (Servlet, JSP, Node.js, MongoDB, etc)**

DESIGN AND ANALYSIS OF ALGORITHM LAB (BCS553)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to:		
CO 1	Implement algorithm to solve problems by iterative approach.	K ₂ , K ₄
CO 2	Implement algorithm to solve problems by divide and conquer approach	K ₃ , K ₅
CO 3	Implement algorithm to solve problems by Greedy algorithm approach.	K ₄ , K ₅
CO 4	Implement algorithm to solve problems by Dynamic programming, backtracking, branch and bound approach.	K ₄ , K ₅
CO 5	Implement algorithm to solve problems by branch and bound approach.	K ₃ , K ₄
DETAILED SYLLABUS		
1. Program for Recursive Binary & Linear Search. 2. Program for Heap Sort. 3. Program for Merge Sort. 4. Program for Selection Sort. 5. Program for Insertion Sort. 6. Program for Quick Sort. 7. Knapsack Problem using Greedy Solution 8. Perform Travelling Salesman Problem 9. Find Minimum Spanning Tree using Kruskal's Algorithm 10. Implement N Queen Problem using Backtracking 11. Sort a given set of n integer elements using Quick Sort method and compute its time complexity. Run the program for varied values of n> 5000 and record the time taken to sort. Plot a graph of the time taken versus non graph sheet. The elements can be read from a file or can be generated using the random number generator. Demonstrate using Java how the divide and- conquer method works along with its time complexity analysis: worst case, average case and best case. 12. Sort a given set of n integer elements using Merge Sort method and compute its time complexity. Run the program for varied values of n> 5000, and record the time taken to sort. Plot a graph of the time taken versus non graph sheet. The elements can be read from a file or can be generated using the random number generator. Demonstrate how the divide and- conquer method works along with its time complexity analysis: worst case, average case and best case. 13.6. Implement , the 0/1 Knapsack problem using (a) Dynamic Programming method (b) Greedy method. 14. From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm. 15. Find Minimum Cost Spanning Tree of a given connected undirected graph using Kruskal's algorithm. Use Union-Find algorithms in your program. 16. Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm. 17. Write programs to (a) Implement All-Pairs Shortest Paths problem using Floyd's algorithm. (b) Implement Travelling Sales Person problem using Dynamic programming. 18. Design and implement to find a subset of a given set $S = \{S_1, S_2, \dots, S_n\}$ of n positive integers whose SUM is equal to a given positive integer d. For example, if $S = \{1, 2, 5, 6, 8\}$ and $d = 9$, there are two solutions $\{1,2,6\}$ and $\{1,8\}$. Display a suitable message, if the given problem instance doesn't have a solution. 19. Design and implement to find all Hamiltonian Cycles in a connected undirected Graph G of n vertices using backtracking principle.		
Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner It is also suggested that open source tools should be preferred to conduct the lab (C, C++ etc)		

Software Engineering (BCS601)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to		
CO 1	Explain various software characteristics and analyze different software Development Models.	K ₁ , K ₂
CO 2	Demonstrate the contents of a SRS and apply basic software quality assurance practices to ensure that design, development meet or exceed applicable standards.	K ₁ , K ₂
CO 3	Compare and contrast various methods for software design	K ₂ , K ₃
CO 4	Formulate testing strategy for software systems, employ techniques such as unit testing, Test driven development and functional testing.	K ₃
CO 5	Manage software development process independently as well as in teams and make use of Various software management tools for development, maintenance and analysis.	K ₅
DETAILED SYLLABUS		3-1-0
Unit	Topic	Proposed Lecture
I	Introduction: Introduction to Software Engineering, Software Components, Software Characteristics, Software Crisis, Software Engineering Processes, Similarity and Differences from Conventional Engineering Processes, Software Quality Attributes. Software Development Life Cycle (SDLC) Models: Water Fall Model, Prototype Model, Spiral Model, Evolutionary Development Models, Iterative Enhancement Models.	08
II	Software Requirement Specifications (SRS): Requirement Engineering Process: Elicitation, Analysis, Documentation, Review and Management of User Needs, Feasibility Study, Information Modelling, Data Flow Diagrams, Entity Relationship Diagrams, Decision Tables, SRS Document, IEEE Standards for SRS. Software Quality Assurance (SQA): Verification and Validation, SQA Plans, Software Quality Frameworks, ISO 9000 Models, SEI-CMM Model.	08
III	Software Design: Basic Concept of Software Design, Architectural Design, Low Level Design: Modularization, Design Structure Charts, Pseudo Codes, Flow Charts, Coupling and Cohesion Measures, Design Strategies: Function Oriented Design, Object Oriented Design, Top-Down and Bottom-Up Design. Software Measurement and Metrics: Various Size Oriented Measures: Halstead's Software Science, Function Point (FP) Based Measures, Cyclomatic Complexity Measures: Control Flow Graphs.	08
IV	Software Testing: Testing Objectives, Unit Testing, Integration Testing, Acceptance Testing, Regression Testing, Testing for Functionality and Testing for Performance, TopDown and Bottom-Up Testing Strategies: Test Drivers and Test Stubs, Structural Testing (White Box Testing), Functional Testing (Black Box Testing), Test Data Suit Preparation, Alpha and Beta Testing of Products. Static Testing Strategies: Formal Technical Reviews (Peer Reviews), Walk Through, Code Inspection, Compliance with Design and Coding Standards.	08
V	Software Maintenance and Software Project Management: Software as an Evolutionary Entity, Need for Maintenance, Categories of Maintenance: Preventive, Corrective and Perfective Maintenance, Cost of Maintenance, Software Re- Engineering, Reverse Engineering. Software Configuration Management Activities, Change Control Process, Software Version Control, An Overview of CASE Tools. Estimation of Various Parameters such as Cost, Efforts,	08

	Schedule/Duration, Constructive Cost Models (COCOMO), Resource Allocation Models, Software Risk Analysis and Management.	
Text books: 1.RS Pressman, Software Engineering: A Practitioners Approach, McGraw Hill. 2. Pankaj Jalote, Software Engineering, Wiley 3. Rajib Mall, Fundamentals of Software Engineering, PHI Publication. 4. KK Aggarwal and Yogesh Singh, Software Engineering, New Age International Publishers. 5. Ghezzi, M. Jarayeri, D. Manodrioli, Fundamentals of Software Engineering, PHI Publication. 6. Ian Sommerville, Software Engineering, Addison Wesley. 7. Kassem Saleh, “Software Engineering”, Cengage Learning. 8. P fleejer, Software Engineering, Macmillan Publication		

Data Science in IOT (BCIT601)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to understand		
CO 1	Explain the fundamentals of data science and its importance.	K ₃
CO 2	Classify the evolution, roles, stages in data science projects	K ₃ , K ₄
CO 3	Analyze the pre-processing and data reduction strategies	K ₂ , K ₃
CO 4	Explain the different data visualization and representation techniques	K ₂ , K ₄
CO 5	Evaluate the performance of algorithms in data science	K ₃ , K ₆
DETAILED SYLLABUS		3-1-0
Unit	Topic	Proposed Lecture
I	Introduction to Data Analytics: Defining IoT Analytics and Challenges: The situation, Defining IoT analytics, IoT analytics challenges, Business value concern, IoT Analytics for the Cloud, Types of Analytics: Streaming Analytics, Spatial, Time Series and Prescriptive Analytics	08
II	Data Collection: Getting to know your data, Types of Data, Data collection strategies, Data Pre engineering with IoT data, Exploratory Data Analytics, Descriptive Statistics, Mean, Deviation, Skewness and Kurtosis	08
III	Data Visualization and Representation: Model Development Simple and Multiple Regression, Model Evaluation using Visualization, Residual Plot, Distribution Plot, Polynomial Regression and Pipelines, Measures for In Evaluation, Prediction and Decision Making, Box Plots, Pivot Table, Heat Map	08
IV	Strategies to Organize Data for Analytics: Linked Analytical Datasets, Linking together strategy, Economics of IoT Analytics, Cost considerations revenue opportunities, The economics of predictive maintenance Cycle	08
V	Application of Analytics in IoT: IoT based applications, Healthcare, Marketing, Finance, Smart surveillance, Agriculture and Weather Forecasting and other domains; Real analysis	08
Text books: 1. JojoMoolayil, “Smarter Decisions : The Intersection of IoT and Data Science”, PACKT, 2016. 2. Cathy O’Neil and Rachel Schu 3. David Dietrich, Barry Heller, Beibei Yang, “Data Science and Big data Analytics”, EMC 2013 4. Andrew Minter, “Analytics for the Internet of Things (IoT)” 2017, ISBN 978-1787120730. 5. HwaiyuGeng, Internet of Things and Data Analytics Handbook (1s ISBN 978-1119173649.		

Computer Networks (BCS603)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to		
CO1	Explain basic concepts, OSI reference model, services and role of each layer of OSI model and TCP/IP, networks devices and transmission media, Analog and digital data transmission	K ₁ ,K ₂
CO2	Apply channel allocation, framing, error and flow control techniques.	K ₃
CO3	Describe the functions of Network Layer i.e. Logical addressing, subnetting & Routing Mechanism.	K ₂ ,K ₃
CO4	Explain the different Transport Layer function i.e. Port addressing, Connection Management, Error control and Flow control mechanism.	K ₂ ,K ₃
CO5	Explain the functions offered by session and presentation layer and their Implementation.	K ₂ ,K ₃
CO6	Explain the different protocols used at application layer i.e. HTTP, SNMP, SMTP, FTP, TELNET and VPN.	K ₂
DETAILED SYLLABUS		3-1-0
Unit	Topic	Proposed Lecture
I	Introductory Concepts: Goals and applications of networks, Categories of networks, Organization of the Internet, ISP, Network structure and architecture (layering principles, services, protocols and standards), The OSI reference model, TCP/IP protocol suite, Network devices and components. Physical Layer: Network topology design, Types of connections, Transmission media, Signal transmission and encoding, Network performance and transmission impairments, Switching techniques and multiplexing.	08
II	Link layer: Framing, Error Detection and Correction, Flow control (Elementary Data Link Protocols, Sliding Window protocols). Medium Access Control and Local Area Networks: Channel allocation, Multiple access protocols, LAN standards, Link layer switches & bridges (learning bridge and spanning tree algorithms).	08
III	Network Layer: Point-to-point networks, Logical addressing, Basic internetworking (IP, CIDR, ARP, RARP, DHCP, ICMP), Routing, forwarding and delivery, Static and dynamic routing, Routing algorithms and protocols, Congestion control algorithms, IPv6.	08
IV	Transport Layer: Process-to-process delivery, Transport layer protocols (UDP and TCP), Multiplexing, Connection management, Flow control and retransmission, Window management, TCP Congestion control, Quality of service.	08
V	Application Layer: Domain Name System, World Wide Web and Hyper Text Transfer Protocol, Electronic mail, File Transfer Protocol, Remote login, Network management, Data compression, Cryptography – basic concepts.	08
Text books and References: 1. Behrouz Forouzan, “Data Communication and Networking”, McGraw Hill 2. Andrew Tanenbaum “Computer Networks”, Prentice Hall. 3. William Stallings, “Data and Computer Communication”, Pearson. 4. Kurose and Ross, “Computer Networking- A Top-Down Approach”, Pearson. 5. Peterson and Davie, “Computer Networks: A Systems Approach”, Morgan Kaufmann 6. W. A. Shay, “Understanding Communications and Networks”, Cengage Learning. 7. D. Comer, “Computer Networks and Internets”, Pearson. 8. Behrouz Forouzan, “TCP/IP Protocol Suite”, McGraw Hill.		

ENERGY HARVESTING TECHNOLOGIES AND POWERMANAGEMENT FOR IOT DEVICES (BCIT 061)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to:		
CO 1	Understand the various energy sources and energy harvesting based sensor networks	K ₁ , K ₂
CO 2	Learn about the various Piezoelectric materials and Non-linear techniques	K ₁ , K ₂
CO 3	Understand the various Power sources for WSN	K ₁ , K ₂
CO 4	Learn about the applications of Energy harvesting systems.	K ₁ , K ₂
DETAILED SYLLABUS		3-0-0
Unit	Topic	Proposed Lecture
I	ENERGY HARVESTING SYSTEMS: Introduction – Energy sources – energy harvesting based sensor networks –photovoltaic cell technologies – generation of electric power in semiconductor PV cells – types	08
II	PIEZO-ELECTRIC ENERGY HARVESTING AND ELECTROMECHANICAL MODELING : Piezoelectric materials – transducers – harvesters – microgenerators – strategies for enhancing the performance of energy harvesters. Electromechanical modeling of Lumped parameter model and coupled distributed parameter models and closed-form Solutions	
III	ELECTROMAGNETIC ENERGY HARVESTING AND NON-LINEAR TECHNIQUES : Basic principles – micro fabricated coils and magnetic materials – scaling – power maximations – micro and macro scale implementations. Non-linear techniques –vibration control & steady state cases	
IV	ENERGY HARVESTING WIRELESS SENSORS: Power sources for WSN – Power generation – conversion – examples – case studies. Harvesting microelectronic circuits – power conditioning and losses	
V	SELECTED APPLICATIONS OF ENERGY HARVESTING SYSTEMS: Case studies for Implanted medical devices – Bio-MEMS based applications – harvesting for RF sensors and ID tags – powering wireless SHM sensor nodes	08
Text Books: 1. Carlos Manuel Ferreira Carvalho, Nuno Filipe Silva VerissimoPaulino, “CMOS Indoor Light Energy Harvesting System for Wireless Sensing Applications”, springer 2. Danick Briand, Eric Yeatman, Shad Roundy ,“Micro Energy Harvesting”		

MACHINE LEARNING TECHNIQUES (BCDS062)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able:		
CO 1	To understand the need for machine learning for various problem solving	K ₁ , K ₂
CO 2	To understand a wide variety of learning algorithms and how to evaluate models generated from data	K ₁ , K ₃
CO 3	To understand the latest trends in machine learning	K ₂ , K ₃
CO 4	To design appropriate machine learning algorithms and apply the algorithms to a real-world problems	K ₄ , K ₆
CO 5	To optimize the models learned and report on the expected accuracy that can be achieved by applying the models	K ₄ , K ₅
DETAILED SYLLABUS		3-0-0
Unit	Topic	Proposed Lecture
I	INTRODUCTION – Learning, Types of Learning, Well defined learning problems, Designing a Learning System, History of ML, Introduction of Machine Learning Approaches – (Artificial Neural Network, Clustering, Reinforcement Learning, Decision Tree Learning, Bayesian networks, Support Vector Machine, Genetic Algorithm), Issues in Machine Learning and Data Science Vs Machine Learning;	08
II	REGRESSION: Linear Regression and Logistic Regression BAYESIAN LEARNING – Bayes theorem, Concept learning, Bayes Optimal Classifier, Naïve Bayes classifier, Bayesian belief networks, EM algorithm. SUPPORT VECTOR MACHINE: Introduction, Types of support vector kernel – (Linear kernel, polynomial kernel, and Gaussian kernel), Hyperplane – (Decision surface), Properties of SVM, and Issues in SVM.	08
III	DECISION TREE LEARNING – Decision tree learning algorithm, Inductive bias, Inductive inference with decision trees, Entropy and information theory, Information gain, ID-3 Algorithm, Issues in Decision tree learning. INSTANCE-BASED LEARNING – k-Nearest Neighbour Learning, Locally Weighted Regression, Radial basis function networks, Case-based learning.	08
IV	ARTIFICIAL NEURAL NETWORKS – Perceptron's, Multilayer perceptron, Gradient descent and the Delta rule, Multilayer networks, Derivation of Backpropagation Algorithm, Generalization, Unsupervised Learning – SOM Algorithm and its variant; DEEP LEARNING –Introduction, concept of convolutional neural network , Types of layers – (Convolutional Layers , Activation function , pooling , fully connected) , Concept of Convolution (1D and 2D) layers, Training of network, Case study of CNN for eg on Diabetic Retinopathy, Building a smart speaker, Self-deriving car etc.	08
V	REINFORCEMENT LEARNING –Introduction to Reinforcement Learning , Learning Task, Example of Reinforcement Learning in Practice, Learning Models for Reinforcement – (Markov Decision process , Q Learning – Q Learning function, Q Learning Algorithm), Application of Reinforcement Learning, Introduction to Deep Q Learning. GENETIC ALGORITHMS: Introduction, Components, GA cycle of reproduction, Crossover, Mutation, Genetic Programming, Models of Evolution and Learning, Applications.	08
Text books: 1. Tom M. Mitchell, —Machine Learning, McGraw-Hill Education (India) Private Limited, 2013. 2. Ethem Alpaydin, —Introduction to Machine Learning (Adaptive Computation and Machine Learning), MIT Press 2004. 3. Stephen Marsland, —Machine Learning: An Algorithmic Perspective, CRC Press, 2009. 4. Bishop, C., Pattern Recognition and Machine Learning. Berlin: Springer-Verlag. 5. M. Gopal, “Applied Machine Learning”, McGraw Hill Education		

Blockchain Architecture Design (BCS063)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to		
CO 1	Describe the basic understanding of Blockchain architecture along with its primitive.	K ₁ , K ₂
CO 2	Explain the requirements for basic protocol along with scalability aspects.	K ₂ , K ₃
CO 3	Design and deploy the consensus process using frontend and backend.	K ₃ , K ₄
CO 4	Apply Blockchain techniques for different use cases like Finance, Trade/Supply and Government activities.	K ₄ , K ₅
DETAILED SYLLABUS		3-0-0
Unit	Topic	Proposed Lecture
I	Introduction to Blockchain: Digital Money to Distributed Ledgers , Design Primitives: Protocols, Security, Consensus, Permissions, Privacy. Blockchain Architecture and Design: Basic crypto primitives: Hash, Signature,) Hashchain to Blockchain, Basic consensus mechanisms	08
II	Consensus: Requirements for the consensus protocols, Proof of Work (PoW), Scalability aspects of Blockchain consensus protocols Permissioned Blockchains: Design goals, Consensus protocols for Permissioned Blockchains	08
III	Hyperledger Fabric (A): Decomposing the consensus process , Hyperledger fabric components, Chaincode Design and Implementation Hyperledger Fabric (B): Beyond Chaincode: fabric SDK and Front End (b) Hyperledger composer tool	08
IV	Use case 1 : Blockchain in Financial Software and Systems (FSS): (i) Settlements, (ii) KYC, (iii) Capital markets, (iv) Insurance Use case 2: Blockchain in trade/supply chain: (i) Provenance of goods, visibility, trade/supply chain finance, invoice management discounting, etc	08
V	Use case 3: Blockchain for Government: (i) Digital identity, land records and other kinds of record keeping between government entities, (ii) public distribution system social welfare systems Blockchain Cryptography, Privacy and Security on Blockchain	08
Text books: 1. Mastering Bitcoin: Unlocking Digital Cryptocurrencies, by Andreas Antonopoulos 2. Blockchain by Melanie Swa, O'Reilly 3. Hyperledger Fabric - https://www.hyperledger.org/projects/fabric 4. Zero to Blockchain - An IBM Redbooks course, by Bob Dill, David Smits - https://www.redbooks.ibm.com/Redbooks.nsf/RedbookAbstracts/crse0401.html		

Data Compression (BCS064)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to:		
CO 1	To gain a fundamental understanding of data compression methods for text, images, and video, and related issues in the storage, access, and use of large data sets	K ₁ , K ₂
CO 2	To select, giving reasons that are sensitive to the specific application and particular circumstance, most appropriate compression techniques for text, audio, image and video information	K ₂ , K ₃
CO 3	To illustrate the concept of various algorithms for compressing text, audio, image and video information.	K ₂ , K ₃
CO 4	To understand various Distortion criterias.	K ₂ , K ₃
CO 5	To illustrate the Advantages of Vector Quantization over Scalar Quantization.	K ₂ , K ₃
DETAILED SYLLABUS		3-0-0
Unit	Topic	Proposed Lecture
I	Compression Techniques: Loss less compression, Lossy Compression, Measures of performance, Modeling and coding, Mathematical Preliminaries for Lossless compression: A brief introduction to information theory, Models: Physical models, Probability models, Markov models, composite source model, Coding: uniquely decodable codes, Prefix codes.	08
II	The Huffman coding algorithm: Minimum variance Huffman codes, Adaptive Huffman coding: Update procedure, Encoding procedure, Decoding procedure. Golomb codes, Rice codes, Tunstall codes, Applications of Hoffman coding: Loss less image compression, Text compression, Audio Compression.	08
III	Coding a sequence, Generating a binary code, Comparison of Binary and Huffman coding, Applications: Bi-level image compression-The JBIG standard, JBIG2, Image compression. Dictionary Techniques: Introduction, Static Dictionary: Diagram Coding, Adaptive Dictionary. The LZ77 Approach, The LZ78 Approach, Applications: File Compression-UNIX compress, Image Compression: The Graphics Interchange Format (GIF), Compression over Modems: V.42 bits, Predictive Coding: Prediction with Partial match (ppm): The basic algorithm, The ESCAPE SYMBOL, length of context, The Exclusion Principle, The Burrows-Wheeler Transform: Move-to-front coding, CALIC, JPEG-LS, Multi-resolution Approaches, Facsimile Encoding, Dynamic Markov Compression.	08
IV	Distortion criteria, Models, Scalar Quantization: The Quantization problem, Uniform Quantizer, Adaptive Quantization, Non uniform Quantization.	08
V	Advantages of Vector Quantization over Scalar Quantization, The Linde-Buzo-Gray Algorithm, Tree structured Vector Quantizers. Structured Vector Quantizers.	08
Text books: 1. Khalid Sayood, Introduction to Data Compression, Morgan Kaufmann Publishers 2. Elements of Data Compression, Drozdek, Cengage Learning 3. Introduction to Data Compression, Second Edition, Khalid Sayood, The Morgan Kaufmann Series 4. Data Compression: The Complete Reference 4th Edition by David Salomon, Springer 5. Text Compression 1st Edition by Timothy C. Bell Prentice Hall		

Software Engineering Lab (BCS651)

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course , the student will be able to

CO 1	Identify ambiguities, inconsistencies and incompleteness from a requirements specification and state functional and non-functional requirement	K ₂ , K ₄
CO 2	Identify different actors and use cases from a given problem statement and draw use case diagram to associate use cases with different types of relationship	K ₃ , K ₅
CO 3	Draw a class diagram after identifying classes and association among them	K ₄ , K ₅
CO 4	Graphically represent various UML diagrams , and associations among them and identify the logical sequence of activities undergoing in a system, and represent them pictorially	K ₄ , K ₅
CO 5	Able to use modern engineering tools for specification, design, implementation and testing	K ₃ , K ₄

DETAILED SYLLABUS

For any given case/ problem statement do the following;

1. Prepare a SRS document in line with the IEEE recommended standards.
2. Draw the use case diagram and specify the role of each of the actors. Also state the precondition, post condition and function of each use case.
3. Draw the activity diagram.
4. Identify the classes. Classify them as weak and strong classes and draw the class diagram.
5. Draw the sequence diagram for any two scenarios.
6. Draw the collaboration diagram.
7. Draw the state chart diagram.
8. Draw the component diagram.
9. Perform forward engineering in java. (Model to code conversion)
10. Perform reverse engineering in java. (Code to Model conversion)
11. Draw the deployment diagram.

Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner
It is also suggested that open source tools should be preferred to conduct the lab (Open Office , Libra , Junit, Open Project , GanttProject , dotProject, AgroUML, StarUML etc.)

INTERNET OF THINGS LAB (BCIT651)

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course , the student will be able to

CO 1	Understanding the implementation of IOT	K ₅ , K ₆
CO 2	Solving Societal problems with the help of IOT	K ₅ , K ₆
CO 3	Problem Analysis and Designing a Solution	K ₅ , K ₆
CO 4	Understanding the importance of Technology in the life of common men.	K ₅ , K ₆

DETAILED SYLLABUS

A students should implement two case studies from the IOT Projects List (Individually or in small group)

1. Wearable Computer With Temperature Distance Sensors
2. Weather Imaging CubeSat with Telemetry Transmission
3. IOT Water Pollution Monitor RC Boat
4. Mountain Climber Health & GPS Tracker
5. Contactless IOT Doorbell
6. IOT Smart Parking Using RFID
7. IOT Contactless Covid Testing Booth Automation
8. IOT Social Distancing & Monitoring Robot For Queue
9. IOT Covid Patient Health Monitor in Quarantine
10. IOT based Manhole Detection and Monitoring System
11. IOT based Smart Energy Meter Monitoring with Theft Detection
12. IOT Weather Station Airship
13. IOT based Three Phase Power Failure Monitoring with SMS Alerts
14. IOT based Intelligent Gas Leakage Detector Using Arduino
15. 360° Aerial Surveillance UAV With IOT Camera
16. IOT Garbage Segregator & Bin Level Indicator
17. IOT Temperature & Mask Scan Entry System
18. IOT based Smart Agriculture Monitoring System Project
19. IOT Based Automatic Vehicle Accident Detection and Rescue System
20. Greenhouse Monitoring and Control System using IOT Project
21. IOT Based Coal Mine Safety Monitoring and Alerting System
22. IOT Based Heart Monitoring System Using ECG
23. IOT based Anti-theft Flooring System using Raspberry Pi
24. Raspberry Pi based Weather Reporting Over IOT
25. IOT Early Flood Detection & Avoidance
26. IOT Garbage Monitoring Using Raspberry Pi
27. IOT Circuit Breaker Project
28. IOT Mining Tracking & Worker Safety Helmet
29. IOT Prison Break Monitoring & Alerting System
30. Raspberry Pi Air and Noise Pollution Monitoring System Over IOT

Note: The Instructor may add/delete/modify/tune experiments

COMPUTER NETWORKS LAB (BCS653)

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course , the student will be able to

CO 1	Simulate different network topologies.	K ₃ , K ₄
CO 2	Implement various framing methods of Data Link Layer.	K ₃ , K ₄
CO 3	Implement various Error and flow control techniques.	K ₃ , K ₄
CO 4	Implement network routing and addressing techniques.	K ₃ , K ₄
CO 5	Implement transport and security mechanisms	K ₃ , K ₄

DETAILED SYLLABUS

1. Implementation of Stop and Wait Protocol and Sliding Window Protocol.
2. Study of Socket Programming and Client – Server model
3. Write a code simulating ARP /RARP protocols.
4. Write a code simulating PING and TRACEROUTE commands
5. Create a socket for HTTP for web page upload and download.
6. Write a program to implement RPC (Remote Procedure Call)
7. Implementation of Subnetting .
8. Applications using TCP Sockets like
 - a. Echo client and echo server b. Chat c. File Transfer
9. Applications using TCP and UDP Sockets like d. DNS e. SNMP f. File Transfer
10. Study of Network simulator (NS).and Simulation of Congestion Control Algorithms using NS
11. Perform a case study about the different routing algorithms to select the network path with its optimum and economical during data transfer. i. Link State routing ii. Flooding iii. Distance vector
12. To learn handling and configuration of networking hardware like RJ-45 connector, CAT-6 cable, crimping tool, etc.
13. Configuration of router, hub, switch etc. (using real devices or simulators)
14. Running and using services/commands like ping, traceroute, nslookup, arp, telnet, ftp, etc.
15. Network packet analysis using tools like Wireshark, tcpdump, etc.
16. Network simulation using tools like Cisco Packet Tracer, NetSim, OMNeT++, NS2, NS3, etc.
17. Socket programming using UDP and TCP (e.g., simple DNS, data & time client/server, echo client/server, iterative & concurrent servers)

Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner

It is also suggested that open source tools should be preferred to conduct the lab (C , C++ , Java , NS3, Mininet, Opnet, TCP Dump, Wireshark etc.